LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can place rational numbers, including fractions and decimals, on a number line.

WHY THIS MATTERS Arihi planted a flower. The height and depth of the plant are shown. How do the height of the plant and the depth of the roots compare?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Rational number *Spanish: Número racional*	A number that can be written as a fraction of two integers, with a bottom number that is not zero.	−3/4, 0.5, 7, −2.85
Fraction *Spanish: Fracción*	A number that shows part of a whole, like 3/4.	3/4 of a cup
Decimal *Spanish: Decimal*	A number with a dot, like 0.5, that shows a part less than one.	0.5 is half of one
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.	left is less, right is more
Equivalent *Spanish: Equivalente*	Having the same value, just written a different way.	1/2 = 0.5
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Plot 1.5 and −3.5 on a number line.

1. Mark the integers from −4 to 4.
2. Split each gap into halves.
3. 1.5 is halfway between 1 and 2. −3.5 is halfway between −4 and −3.

Answer: 1.5 is right of 0; −3.5 is left of 0

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's plot 0.75. Which two integers is it between? It is between 0 and 1.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The number _____ is rational because it can be written as _____ .
- On the number line, _____ falls between _____ and _____ .
- I split each unit into _____ equal parts because _____ .
- The opposite of _____ is _____ , so it sits to the _____ of zero.

Word bank rational number | fraction | decimal | opposite | number line | integer | between | equal parts | denominator | above | below | zero

WATCH OUT!

A common mistake

A common mistake in Rational Numbers on the Number Line is treating the numerator and denominator of a negative fraction as two separate integers instead of one value between two integers

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each fraction to its decimal equivalent.

_____	1/2	A	0.5
_____	-3/4	B	-0.75
_____	1 1/4	C	1.25
_____	-2 1/2	D	-2.5
_____	3/10	E	0.3
_____	-1/5	F	-0.2

STEPS TO TRY

1. Each fraction equals a decimal. Halves, quarters, fifths, and tenths all have decimal partners you know.
2. Divide top by bottom and keep the sign: -3/4 stays negative. For mixed numbers, keep the whole part and convert the fraction.
3. Start with 1/2: 1 ÷ 2 = 0.5 — match those two. Now convert 3/10 by reading it as 'three tenths.'

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which point is located between -1 and -2 on the number line?

(A) -2.5
(B) -1.5
(C) -0.5
(D) 1.5

STEPS TO TRY

1. 'Between -1 and -2' is on the NEGATIVE side, in the gap from -1 to -2.
2. Turn each choice into a spot on the number line and check: is it past -1 but not past -2?
3. Start at -1 and move half a unit farther left. Which choice lands in that gap between -1 and -2?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which fraction is equivalent to 0.25?

(A) 1/4
(B) 1/2
(C) 1/3
(D) 2/5

STEPS TO TRY

1. Read 0.25 as a place-value number: 25 hundredths.
2. Write it as 25/100, then simplify by dividing the top and bottom by the same number.
3. Divide both 25 and 100 by 25. What simpler fraction do you get?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Where is -3/4 located on the number line?

(A) Between -3 and -4

(B) Between -1 and -2

(C) Between 0 and -1

(D) Between 0 and 1

STEPS TO TRY

1. The size of -3/4 is less than 1 whole — its distance from zero is under 1.
2. Convert: 3/4 = 0.75, so -3/4 = -0.75. Now ask which two numbers -0.75 sits between on the line.
3. Start at 0 and move 0.75 units LEFT. Did you pass -1 on the way, or not?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide exactly where to place rational numbers that fall between integers?

SENTENCE STARTER To plot ___, I found the integers it falls between, which are ___ and ___. Then I estimated ___ of the way between them because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help